

Experiment No. 01

Experiment Name: Simulation and Verification of Basic Pneumatic Logic Operations Using FluidSim Software.

Objectives:

- To understand the basic working environment, components, and simulation procedure of FluidSIM software.
- To study the working principle of pneumatic logic valves used to realize Boolean logic operations.
- To verify the output response of each simulated circuit against its corresponding truth table.

Introduction:

FluidSIM is a specialized simulation software, developed by Festo Didactic in cooperation with the Universities of Paderborn and Kaiserslautern that is used for the design, simulation, and analysis of pneumatic, hydraulic, and electro-pneumatic circuits. The software provides a graphical drag-and-drop environment in which standard ISO-symbol components, including compressors, directional control valves, cylinders, sensors, and logic valves, can be selected, interconnected, and animated to observe the dynamic behavior of a circuit under simulated operating conditions.

Pneumatic systems form one of the oldest and most widely used branches of automation technology, in which compressed air is employed as the working medium to transmit energy and signals for the performance of mechanical work. Long before the advent of electronic and digital controllers, pneumatic components such as directional control valves, cylinders, and logic valves were combined to build control circuits capable of performing switching and decision-making functions analogous to those carried out by electronic logic gates. The fundamental Boolean operations, namely AND, OR, NOT, NAND, NOR, and Exclusive-OR (X-OR), can each be physically realized through an appropriate arrangement of pneumatic valves, and such circuits remain relevant in industrial environments where explosion hazards, electromagnetic interference, moisture, or extreme temperatures make electrical control systems unsuitable.

In this report I have discussed and displayed how FluidSIM was employed to design and simulate pneumatic circuits corresponding to the AND, OR, NOT, NAND, NOR, and X-OR logic operations, and the response of each circuit was verified against its respective truth table.

Building Logic Gates:

1. OR Gate:

The OR logic gate was constructed in FluidSIM by using two pneumatic input lines connected to a shuttle valve (OR valve). The two inputs were designated as A and B, while the output was connected to a pneumatic actuator or indicator. Compressed air was supplied to both input lines through separate 3/2-way push-button valves. The outputs of these valves were connected to the two input ports of the shuttle valve. When either input valve

was actuated, compressed air was allowed to pass through the shuttle valve to the output. The same response was obtained when both input valves were actuated simultaneously. Thus, the output became active when A or B or both were active, satisfying the Boolean expression $Y = A + B$.

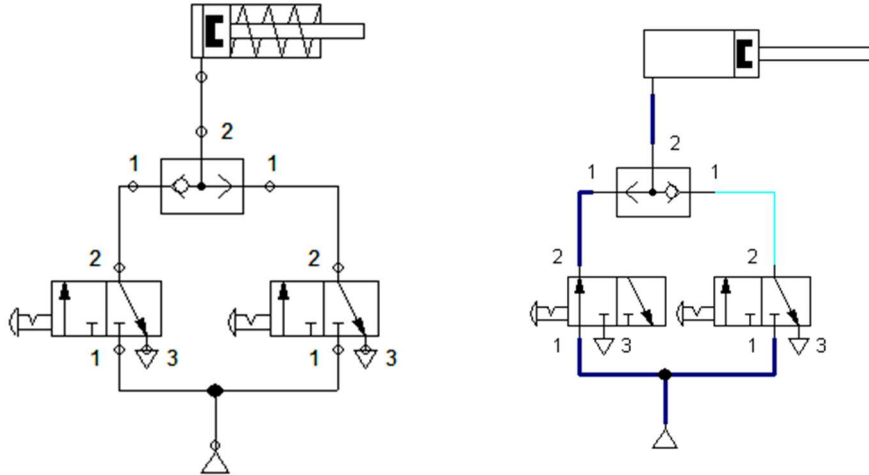
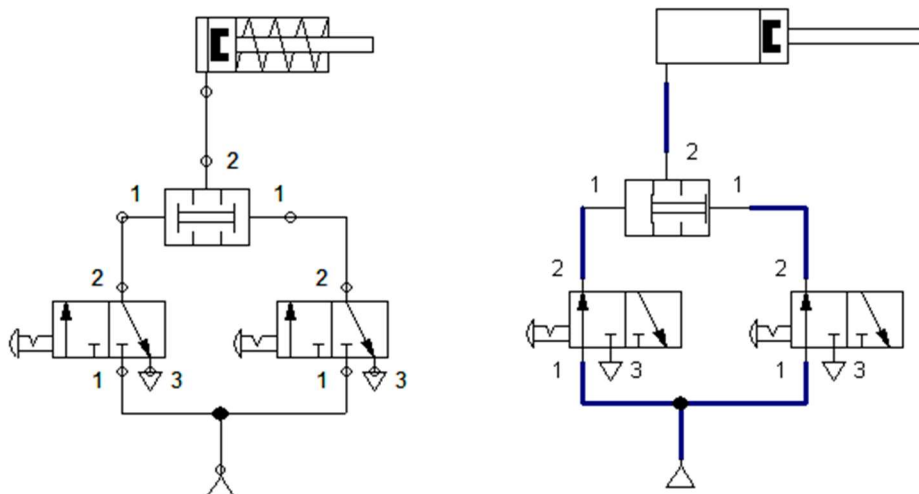


Figure-01: OR gate.

2. AND Gate:

The AND logic gate was constructed by connecting two pneumatic input valves in such a way that compressed air had to pass through both valves before reaching the output. Two 3/2-way push-button valves were used as inputs A and B, and their pneumatic connections were arranged in series. The output of the first valve was connected to the input of the second valve, and the output of the second valve was connected to the actuator or indicator. Air could reach the output only when both input valves were actuated at the same time. If either one of the input valves remained unactuated, the air path was interrupted and no output was produced. Therefore, the circuit performed the AND operation according to $Y = A \cdot B$.



3. NOT Gate:

The NOT logic gate was constructed using a pneumatic valve with a normally open or normally closed configuration arranged so that the output condition became opposite to the input condition. A 3/2-way pneumatic valve was used as the input element, and its actuation was controlled by input A. In the normal state, compressed air was allowed to reach the output, producing an active output when the input was inactive. When input A was actuated, the valve changed its position and interrupted the air supply to the output. Consequently, the output became inactive when the input was active and vice versa. Hence, the circuit performed the NOT operation, represented by $Y = \bar{A}$.

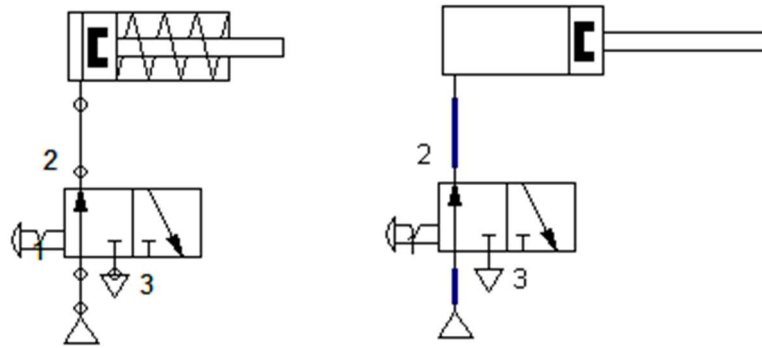
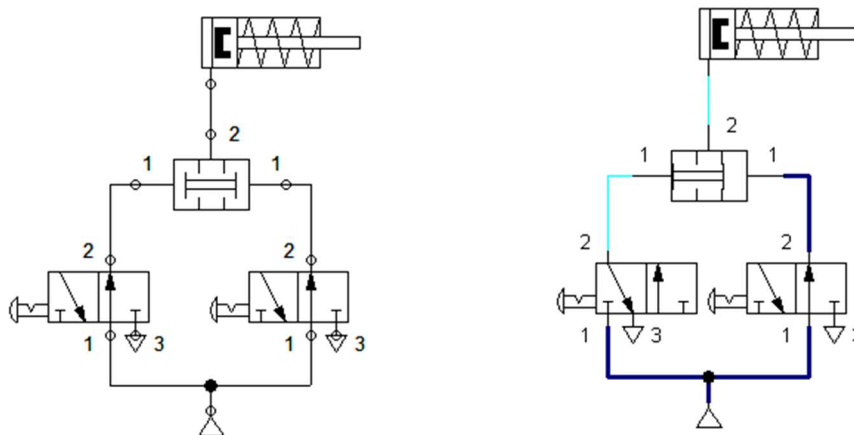


Figure-03: NOT Gate.

4. NOR Gate:

The NOR logic gate was constructed by combining an OR pneumatic arrangement with a NOT function. Two 3/2-way push-button valves were used as inputs A and B, and their outputs were connected to a shuttle valve to perform the OR operation. The output of the shuttle valve was then directed to a pneumatic valve arranged to invert the signal. When either A or B was actuated, an output signal was generated by the OR section, but this signal was subsequently blocked or inverted by the NOT section. Therefore, the final output remained active only when both inputs were inactive. The resulting circuit represented the NOR operation according to $Y = \neg(A + B)$.



5. NAND Gate:

The NAND logic gate was constructed by combining an AND pneumatic arrangement with a NOT function. Two 3/2-way input valves were connected in series so that the output of the AND section became active only when both A and B were actuated. This output was then connected to an inverting pneumatic valve. When both input valves were actuated, the AND section produced an air signal, which caused the inverting section to switch and make the final output inactive. If either one or both inputs were not actuated, the AND output was absent and the final output remained active. Therefore, the circuit performed the NAND operation expressed as $Y = \neg(A \cdot B)$.

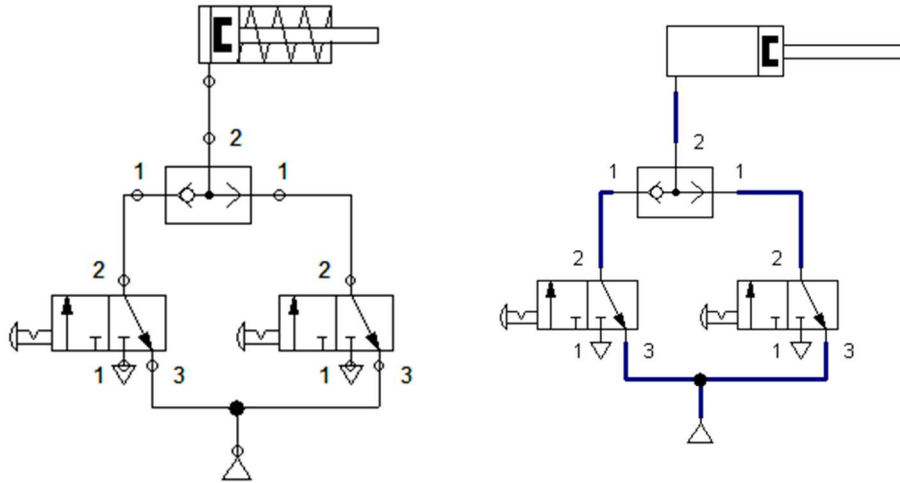


Figure-05: NAND Gate.

6. X-OR Gate:

The XOR logic gate was constructed by combining pneumatic AND, OR, and NOT functions so that the output became active only when one of the two inputs was active. Two input signals A and B were supplied to the pneumatic circuit. An OR arrangement was used to produce an output when either input was active, while an AND arrangement was used to detect the condition in which both inputs were active. The AND signal was then inverted and combined appropriately with the OR signal. As a result, the final output was activated when $A = 1, B = 0$ or $A = 0, B = 1$, but it remained inactive when both inputs were either active or inactive. Thus, the XOR operation was obtained according to $Y = A \oplus B = \bar{A}B + A\bar{B}$.

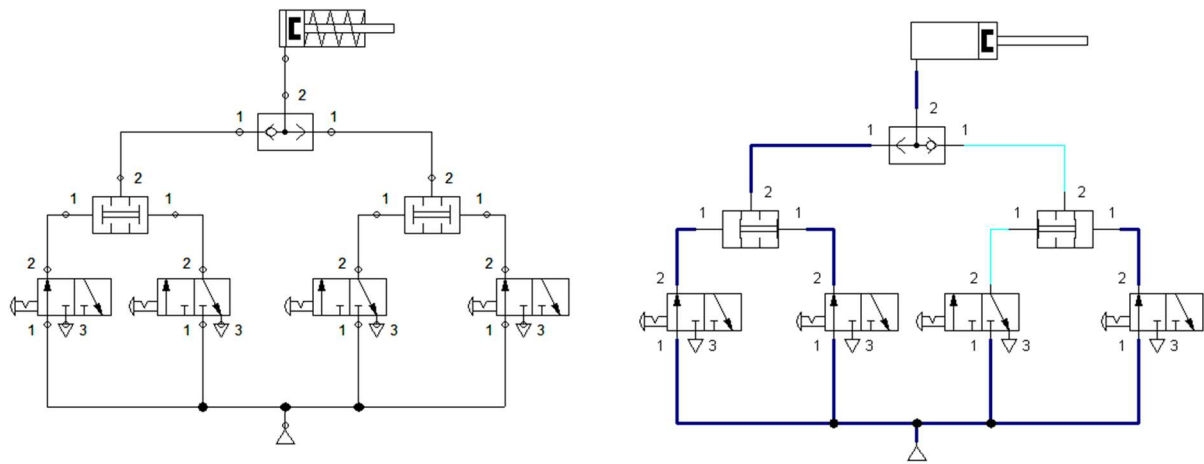


Figure-06: X-OR Gate.

Discussion:

Prior to the implementation of the individual logic circuits, the FluidSIM environment have been introduced, and the basic components required for pneumatic circuit construction, including the compressed air supply, push-button-actuated three by two directional control valves used as input signal generators, two-pressure valves, shuttle valves, and single-acting cylinders used as output indicators, have been identified and placed on the simulation worksheet. Connections between components have been made using pneumatic supply lines, and each circuit have been subsequently simulated by activating the input valves in the required combinations while the resulting extension or retraction of the output cylinder was observed and recorded. Some problems have been occurred during the simulation that, the NOR, NAND and X-OR Gate gave inconsistent result while $A=0, B=0$ & $A=1, B=1$ conditions. So, there have been some limitation designing these three circuits.

Conclusion:

Experimental results have been significantly matched with the theoretical result. Some problem and design limitations have also been recognized. The introduction and importance of pneumatic control have been understood and the use of FluidSim in this field have been realized. To sum up, through this exercise, a foundational understanding of pneumatic logic control have been gained, which is expected to be of value in subsequent studies of electro-pneumatic and industrial automation systems.

References:

- [1] Festo Didactic GmbH & Co. KG, FluidSIM 5 Pneumatics User Manual, Festo Didactic, Denkendorf, Germany.
- [2] Croser, P. and Ebel, F., Pneumatics: Basic Level, Festo Didactic GmbH & Co. KG, Esslingen, Germany.
- [3] Majumdar, S. R., Pneumatic Systems: Principles and Maintenance, Tata McGraw-Hill Education, New Delhi, India.